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Advancing Diabetic Retinopathy Detection: Leveraging Deep Learning for Accurate Classification and Early Diagnosis

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Aakash Dilip Kolte
Department of Electronics and Communication Engineering, Faculty of Engineering and Technology SRM Institute of Science and Technology,
Kattankulathur-603203, Tamil Nadu, India
ak4084@srmist.edu.in

Jyotiraditya Sharma
Department of Electronics and Communication Engineering, Faculty of Engineering and Technology SRM Institute of Science and Technology,
Kattankulathur-603203, Tamil Nadu, India
js2099@srmist.edu.in

Utkarsh Rai
Department of Electronics and Communication Engineering, Faculty of Engineering and Technology SRM Institute of Science and Technology,
Kattankulathur-603203, Tamil Nadu, India
ur4767@srmist.edu.in

.

R.Jansi
Department of Electronics and Communication Engineering, Faculty of Engineering and Technology SRM Institute of Science and Technology,
Kattankulathur-603203, Tamil Nadu, India
jansir@srmist.edu.in

***Abstract*—DR is a prevalent diabetes-related complication that can result in vision impairment if not identified and addressed promptly. However, the limited availability of large-scale annotated datasets and the high inter-observer variability of manual diagnosis by ophthalmologists pose challenges for automated DR detection. To achieve the objective, the proposed method involved incorporating transfer learning into a deep learning model, which can improve the accuracy of the model while reducing the amount of training data required. The method involved three main steps: image resizing as preprocessing, feature extraction using Haralick texture features and Hu moments, and classification using various machine learning algorithms including K-NN, Naive Bayes, decision tree, and Support Vector Machine (SVM). The evaluation of the proposed method was conducted on a publicly available Kaggle dataset. The results showed that SVM outperformed the other machine learning algorithms, achieving an accuracy of 52.75%. Furthermore, a comparison was made with an Efficient-Net deep learning model, and the outcomes demonstrated that the deep learning technique achieved the highest accuracy of 91.8% and AUC of 0.91, surpassing all other techniques. This suggests that deep learning techniques, such as Efficient-Net, are well-suited for the detection of diabetic retinopathy.**

Keywords— *Diabetic retinopathy, Feature extraction, Machine learning, Deep learning, Efficient-Net, Accuracy*

I. Introduction

Diabetes has emerged as a prevalent disease in recent years, affecting approximately 382 million people globally, with projections estimating an increase to 592 million by 2025 [1]. There are two types of diabetes, type I and type II, which can result in complications in crucial organs such as the eyes. Diabetic retinopathy (DR) is a significant eye disease associated with diabetes, which can lead to damage of the blood vessels in the retina [2]. Among the 382 million individuals with diabetes, 34.6% have diabetic retinopathy [3]. In the Khyber Pakhtunkhwa (KP) province of Pakistan, DR causes blindness in 4% of the population, with type I diabetes being the most prevalent cause. In KP, 30% of the population has diabetes, and 1.6% of them have type II diabetes [4]. Approximately 30% of diabetic patients have DR, and nearly 2% of these patients have developed complete blindness. The timely detection of DR is crucial in preventing or minimizing its severe consequences [5]. Diabetes impacts the retina, which consists of sensitive nerve cells [6]. DR can be broadly categorized based on the level with which it affects the retina [7]. NPDR can cause severe deterioration of the health of diabetic patients [8]. Initial symptoms include the appearance of tiny, round, and red-colored spots known as microaneurysms (MAs). Hemorrhages (H) can result from weakened blood vessels or MAs breaking and releasing blood, and they can appear as either dot-like or blot-like red lesions. Hemorrhages and MAs are sometimes grouped together and referred to as HMA [9]. Detecting DR in its early stages can be challenging, as it is often asymptomatic. This study aims to leverage advanced deep learning techniques for accurate classification and early diagnosis of DR, with a particular focus on the Efficient-Net model[10]. Developing a robust and efficient DR detection system can contribute to the early management and prevention of vision loss in diabetic patients.

II. Literature Survey

Tsiknakis et al. [11]presented a comprehensive analysis of the application of deep learning techniques for diabetic retinopathy classification using fundus images. The analysis covered various aspects of the detection process, including the commonly used datasets, preprocessing methods that enhanced the performance of models, and lesion localization. The article also discussed models that were implemented in real clinical settings and provided insights and future research directions. Luo et al. [12] proposed a new convolutional

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network for detecting diabetic retinopathy using multiple fundus images. Unlike previous methods that only used a single view, their method took advantage of lesion features from a wider field-of-view (120-150 degrees) and applied attention mechanisms to focus on important views and channels for improved performance. They also assigned higher weights to important channels to enhance feature extraction. Vives-Boix et al. [13] proposed a bio-inspired approach that incorporated synaptic meta-plasticity concept. This biological phenomenon enhances classification capability of the system.

Pires et al. [14] developed a solution based on deep learning and tested the effectiveness of different improvements. This scheme attained an AUC of 98.2% on the Messidor-2 dataset when trained on the Kaggle competition dataset, demonstrating its ability to generalize. Alyoubi et al. [15] proposed that manual diagnosis of diabetic retinopathy is an effortful, and costly process that may result in misdiagnosis. Convolutional neural networks were found to be particularly effective in this regard. Tymchenko et al. [16] introduced a scheme where multiple transfer learning models were coupled together. The proposed method also ranked high, at 54 out of 2943 competing methods, on the APTOS 2019 Blindness Detection Dataset, demonstrating its effectiveness for diabetic retinopathy detection.

Qummar et al. [17] utilized a retina image dataset from Kaggle with the aim of enhancing the classification for various stages of DR by encoding rich features. The findings indicated that the proposed model could detect all stages of DR, unlike existing methods, and outperformed state-of-the-art techniques on the same Kaggle dataset. Khalifa et al. [18] focused on investigating the effectiveness of deep learning for detecting medical diabetic retinopathy using the newly published dataset. The study was considered one of the first to use this dataset. Qiao et al. [19] proposed a system based on CNN. The system was accelerated using GPU based on the common semantic of their image pixels. Khan et al. [20] proposed a study to classify different stages of DR with a deep learning model that had low convergence rate. The SPP layer allowed the model to process images at any scale, while the NiN layer added extra nonlinearity for improved classification.

A.B. Aujih et. al [21] explored the use of U-Net model for retinal vessel segmentation and diabetic retinopathy (DR) severity classification. The presence of retinal vessels was found to be a good feature in classifying DR severity. The study used the MESSIDOR dataset for training and DRIVE dataset for U-Net training. Pre-trained InceptionV1 network was used for DR classification. Jyostna et. al [22] for the classification of diabetic retinopathy severity levels using features extracted from pre-trained VGG-16 convolution blocks. The proposed aggregation module effectively extracted and combined discriminative features from multiple convolution layers of VGG-16. The model achieved state-of-the-art accuracy of 84.31% and an AUC of 97 on the Kaggle APTOS 2019 contest dataset. The main merit of this work is the use of pre-trained models to obtain effective feature representations, while the primary demerit is the limited data used for training.

III. Proposed Methodology

The present study aims to develop a classification model for identifying the presence of diabetic retinopathy (DR) in fundus images. To accomplish this, a range of techniques including preprocessing, feature extraction and classification have been employed, as depicted in Figure 1.

A. Preprocessing (Resize)

Resizing images to a uniform size of 128x128 is a widely used technique in computer vision, particularly in image classification tasks. The original size of the images, typically 224x224, is adjusted to the uniform size of 128x128 for several reasons. Firstly, it reduces memory consumption during feature extraction and classification, which allows the algorithm to handle larger datasets more efficiently. Secondly, it facilitates better detection of patterns and features in the images since all the images are of the same size. This can lead to an improvement in the accuracy of the algorithm. However, there may be a loss of image quality when resizing the images. Nonetheless, the proposed work shows a significant increase in accuracy, which justifies the use of this technique. Overall, resizing images to a fixed size is essential for efficient training and inference in image classification models.

B. Feature Extraction:

Feature extraction is a critical step in which relevant information is extracted from raw data for subsequent analysis. In image recognition, features are extracted from images to represent them in a way suitable for machine learning algorithms. These features include color histograms, texture patterns, and edges. Color histograms reflect the frequency of various colors in an image. Texture patterns capture the repetitive nature of textures within an image, while edges represent the boundaries between different objects or regions in an image. By extracting these features, machine learning algorithms can better differentiate between different objects and patterns in an image, resulting in more accurate results.

Two techniques, Haralick texture features and Hu moments, were used to extract image features from grayscale images in this work. Haralick texture features are statistical measures that describe an image's texture based on the gray-level co-occurrence matrix (GLCM) of the image. This matrix captures the frequency of co-occurring pairs of gray levels. For each image, we extracted 11 Haralick texture features such as contrast, variance, and entropy, among others.

Hu moments, on the other hand, are a set of invariant moments used in image recognition tasks. These moments are derived from the central moments of an image that describe the distribution of pixel intensities around the image's centroid. Seven Hu moments, including measures of rotation invariance, skewness, and symmetry, among others, were extracted for each image. The Haralick texture features and Hu moments were concatenated into a single feature vector for each image, which was then used as input to the machine learning classifiers.

C. Classification

The classification of diabetic retinopathy is a significant task that can be accomplished through the application of classification algorithms. In this study, several classifiers have been employed to classify diabetic retinopathy based on

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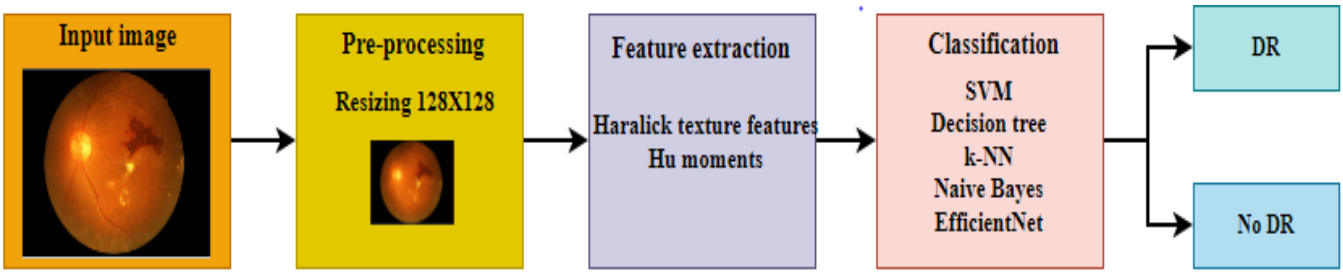

Figure 1. Block diagram of the proposed diabetic retinopathy detection scheme

retinal images that have been resized to a uniform size of 128x128. The classifiers utilized in this research are Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (K-NN), Naïve Bayes, and Efficient-Net. Each classifier has its own strengths and weaknesses, and their performance in classifying diabetic retinopathy has been evaluated using various metrics.

D. Dataset Description

The Diabetic Retinopathy Dataset, available on Kaggle, consists of retinal images of diabetic patients that can aid in the diagnosis of diabetic retinopathy. This dataset comprises 1,960 retinal images with a resolution of 224x224 pixels, grouped into five classes ranging from 0 to 4, reflecting the severity of the diabetic retinopathy. Class 0 represents the absence of diabetic retinopathy, while class 4 denotes proliferative diabetic retinopathy.

This dataset holds potential for various purposes, including the development of machine learning algorithms to diagnose diabetic retinopathy and research on the progression of diabetic retinopathy. However, researchers and practitioners should be aware of the dataset's limitations, such as its relatively small sample size and potential biases in image selection. Consequently, the dataset should be used with caution in research or diagnostic applications.

IV. Performance Evaluation

The classifiers utilized in the study include SVM, Decision Tree, K-NN, Naïve Bayes, and Efficient-Net as shown in Table 1. Images from the Kaggle dataset (https://www.kaggle.com/datasets/sachinkumar413/diabetic-retinopathy-dataset) were used, with 1000 images for DR and 1000 for no DR (including mild DR, moderate DR, proliferate DR, and severe DR). The performance metrics were computed based on [23]. The SVM classifier achieved an accuracy of 52.75% in classifying diabetic retinopathy. The Decision Tree classifier achieved an accuracy of 52% in our experiments, and it is a popular choice for its simplicity of implementation and training-free requirement. The K-NN classifier achieved an accuracy of 48.75% in classifying diabetic retinopathy. Naïve Bayes is a fast, simple, and probabilistic algorithm that assumes independence between features, making it a popular choice for large datasets. Our experiments showed that the Naïve Bayes classifier achieved an accuracy of 49.5% in classifying diabetic retinopathy. Efficient-Net, a deep learning model was fine-tuned to classify diabetic retinopathy in this study, and it outperformed the other classifiers, achieving an accuracy of 91.8%. Our study's results suggest that Efficient-Net is the most effective classifier for diabetic retinopathy classification.

TABLE I. Comparison of Performance Metrics

Classifiers	Accuracy (%)	Recall/Sensitivity	F-score	Specificity (%)
SVM	52.75	63.68	57.52	41.70
Decision Tree	52.00	35.82	42.85	68.34
K-NN	48.75	51.24	50.12	46.23
Naïve Bayes	49.50	41.29	45.10	57.78
Efficient-Net	91.80	90.00	90.90	90.20

Figures 2 and 3 present a comparison of the F-score and recall, respectively, for the classifiers used in this study. The SVM classifier achieved the highest recall/sensitivity score of 63.68%, indicating its ability to correctly identify positive samples. The Decision Tree classifier had a relatively low recall score of 35.82%, implying that it misses many positive samples. The SVM classifier also achieved the highest F-score of 57.52%, taking into account both precision and recall. K-NN and Naïve Bayes achieved similar F-scores of approximately 50%, while Decision Tree had a relatively low F-score of 42.85%. The specificity score measures the ability of the classifier to identify negative samples accurately, and the Decision Tree classifier had the highest specificity score of 68.34%, followed by Naïve Bayes at 57.78%. SVM and K-NN had relatively low specificity scores, indicating their difficulty in identifying negative samples correctly. The Efficient-Net classifier achieved the highest accuracy score of 91.80%, whereas the other classifiers performed significantly worse.

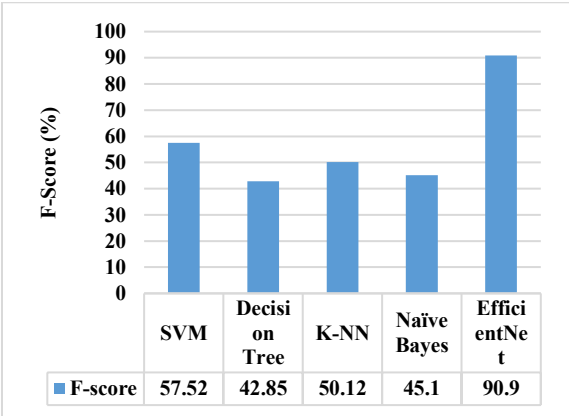

Fig. 2. Comparison of F-score

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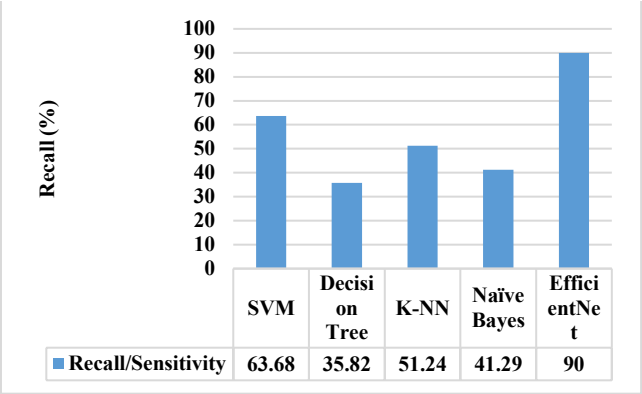

Fig. 3. Comparison of recall

The ROC curves and AUC values for the classifiers are compared in Figures 4 and 5. The SVM classifier had a higher AUC value than the other classifiers, indicating its better performance.

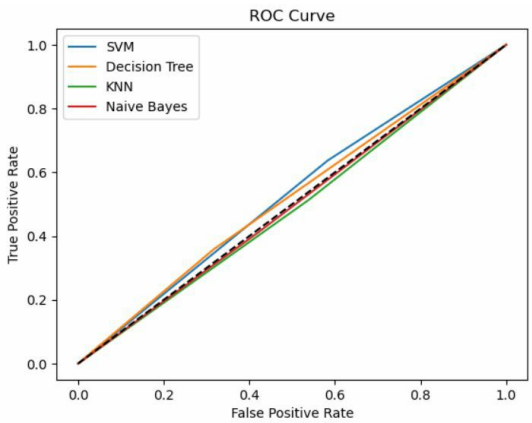

Fig. 4. ROC for machine learning algorithms

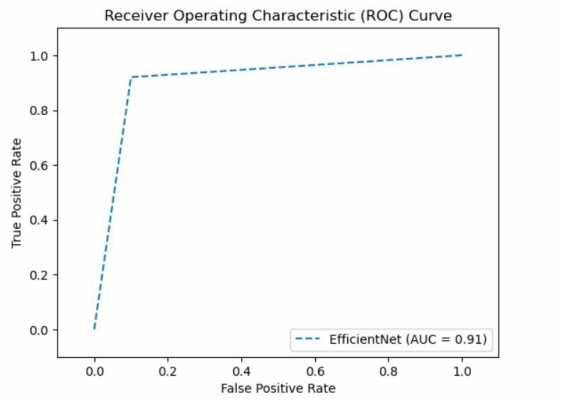

Fig. 5. ROC for Efficient-Net algorithm

Additionally, the study has revealed that the AUC value for Efficient-Net outperforms the AUC values of all the other classifiers, with a value of 0.91. This finding suggests that in this study, Efficient-Net is the most effective classifier for diabetic retinopathy classification. One possible explanation for this result is that Efficient-Net is a deep learning model that has demonstrated exceptional performance in image classification tasks, rendering it a suitable option for medical image analysis tasks.

V. CONCLUSION

In conclusion, the proposed method utilizing preprocessing, feature extraction, and machine learning techniques, with a comparison to deep learning models, provides a promising approach for the detection of diabetic retinopathy. Among the evaluated models, the deep learning technique, specifically 0the Efficient-Net model, demonstrated superior performance with an impressive F-score of 90.90. This highlights the effectiveness of deep learning in accurately classifying diabetic retinopathy, outperforming other classifiers such as SVM (F-score of 57.52), K-NN (F-score of 50.12), Naïve Bayes (F-score of 45.10), and Decision Tree (F-score of 42.85). Further research can be conducted to explore other deep learning models or combinations of techniques to further improve the accuracy and efficiency of diabetic retinopathy detection. This scheme can be used for early diagnosis and management of diabetic retinopathy, which can potentially help lower the likelihood of losing eyesight in diabetic patients.

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